

Based on your **final training logs**, here is a consolidated summary of the MobileNetV3 training results.

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Authentix Video AI Model - Final Training Results

Dataset Summary

Metric	Value
Model	MobileNetV3-Small (ImageNet Pretrained)
Task	Binary Deepfake Detection
Classes	Real / Fake
Total Training Images	40,196
Total Validation Images	7,789
Real Images (Train)	20,498 (51.0%)
Fake Images (Train)	19,698 (49.0%)
Dataset Balance	Nearly Balanced (51:49)
Image Resolution	224 × 224 RGB
Transfer Learning	Yes

Training Configuration

Parameter	Value
Optimizer	AdamW
Loss Function	CrossEntropyLoss + Label Smoothing
Mixed Precision	☒ Enabled (AMP)
Transfer Learning	3 Phases
Early Stopping	Patience = 5

Parameter	Value
Total Parameters	1,075,234
Phase 1 Trainable Params	148,226
Phase 2 Trainable Params	792,866
Phase 3 Trainable Params	1,075,234

Training Phases

Phase	Description	Epochs
Phase 1	Train classifier head only	1-5
Phase 2	Fine-tune last MobileNetV3 blocks	6-15
Phase 3	Fine-tune entire network	16

Best Validation Performance

Metric	Best Value	Epoch
Validation Accuracy	78.28%	11
Validation AUC	0.8737	7
Validation F1 Score	0.6708	11
Best Validation Loss	0.5294	2
Best Model Saved	☒ Yes	Epoch 11 (updated checkpoint)

Note: Different metrics peaked at different epochs. The training log shows the checkpoint being updated later based on the configured selection criterion.

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Final Training Performance

Metric	Value
Final Training Accuracy	95.34%
Final Training Loss	0.2930
Final Validation Accuracy	76.80%
Final Validation Loss	0.6268
Final Validation AUC	0.8547
Final Validation F1	0.6310

Performance Improvements

Metric	Initial Run	Final Run	Improvement
Best Validation Accuracy	~74.0%	78.3%	+4.3%
Best AUC	~0.832	0.874	+0.042
Generalization	Moderate	Better	Improved
GPU Monitoring	Incorrect	Accurate	Fixed
Training Stability	Basic	Improved	Better

Training Speed

Metric	Value
Fastest Training Speed	1455.9 images/sec
Average Training Speed	~1250-1400 images/sec
Average Validation Time	~3-4 sec

Metric	Value
Average Training Epoch Time	~28-33 sec
Phase 3 Epoch Time	~46 sec

GPU Resource Usage

Metric	Value
Peak VRAM Allocated	306.5 MB
Peak VRAM Reserved	352.0 MB
Device	NVIDIA GPU (CUDA)
AMP	Enabled

Regularization Used

Technique	Status
Label Smoothing	✓
Weight Decay	✓
Dropout	✓
Balanced Class Weights	✓
Early Stopping	✓
Mixed Precision	✓
Transfer Learning	✓

Generated Outputs

Artifact	Status
best_model.pth	✓
last_model.pth	✓
TensorBoard Logs	✓
Loss Curves	✓
Training Logs	✓
Automatic Best Model Restore	✓

Observations

Observation	Status
Training pipeline executes successfully	✓
Dataset loading works correctly	✓
GPU acceleration functioning	✓
Transfer learning phases executed correctly	✓
Early stopping triggered correctly	✓
Best checkpoint restored automatically	✓
Validation accuracy improved over previous run	✓
Some <code>Grad Norm</code> values reported as <code>inf</code> / <code>nan</code>	⚠ Needs investigation
Mild overfitting remains (95.3% train vs 78.3% best validation accuracy)	⚠ Acceptable but can be improved

Overall Project Status

Component	Status
Dataset Preparation	✔ Complete
Face Extraction Pipeline	✔ Complete
Dataset Balancing	✔ Complete
MobileNetV3 Architecture	✔ Complete
Transfer Learning Pipeline	✔ Complete
Training Pipeline	✔ Complete
Validation Pipeline	✔ Complete
Checkpoint Management	✔ Complete
GPU Optimization	✔ Complete
Model Ready for Inference	✔ Yes

Overall Assessment

Category	Rating
Dataset Quality	★★★★★★
Code Architecture	★★★★★★
Training Pipeline	★★★★★★
Transfer Learning	★★★★★★
Performance	★★★★★☆
Generalization	★★★★★☆
Production Readiness	★★★★★☆

Final Score

Overall Video AI Module: 9.3 / 10

This is a strong milestone for Authentix. The video branch now has a functioning end-to-end training pipeline with a fine-tuned MobileNetV3 model, balanced data, automated checkpointing, GPU acceleration, and improved validation performance. The next logical milestone is to validate the model on the untouched test set and then begin developing the audio branch for multimodal fusion.

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